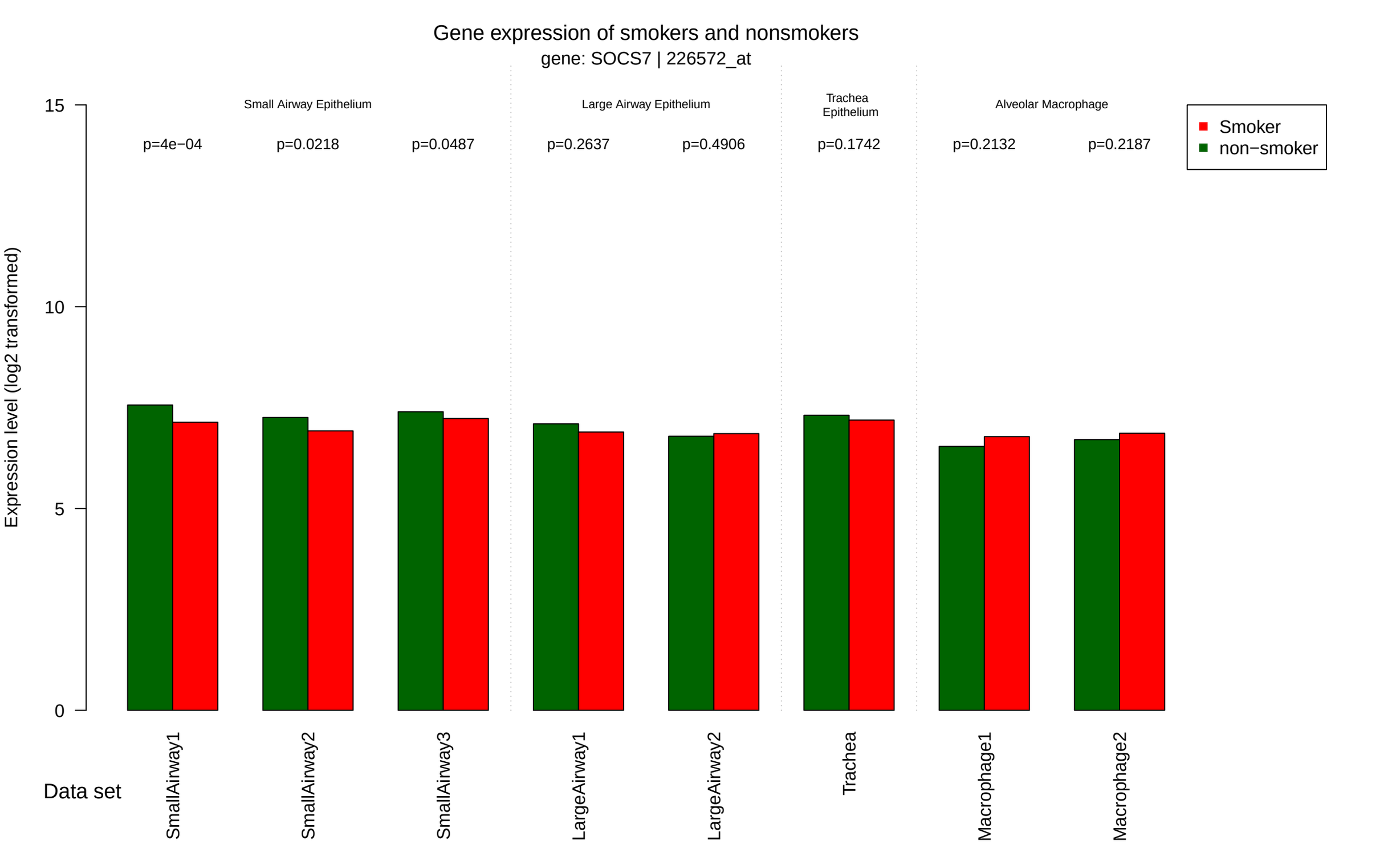
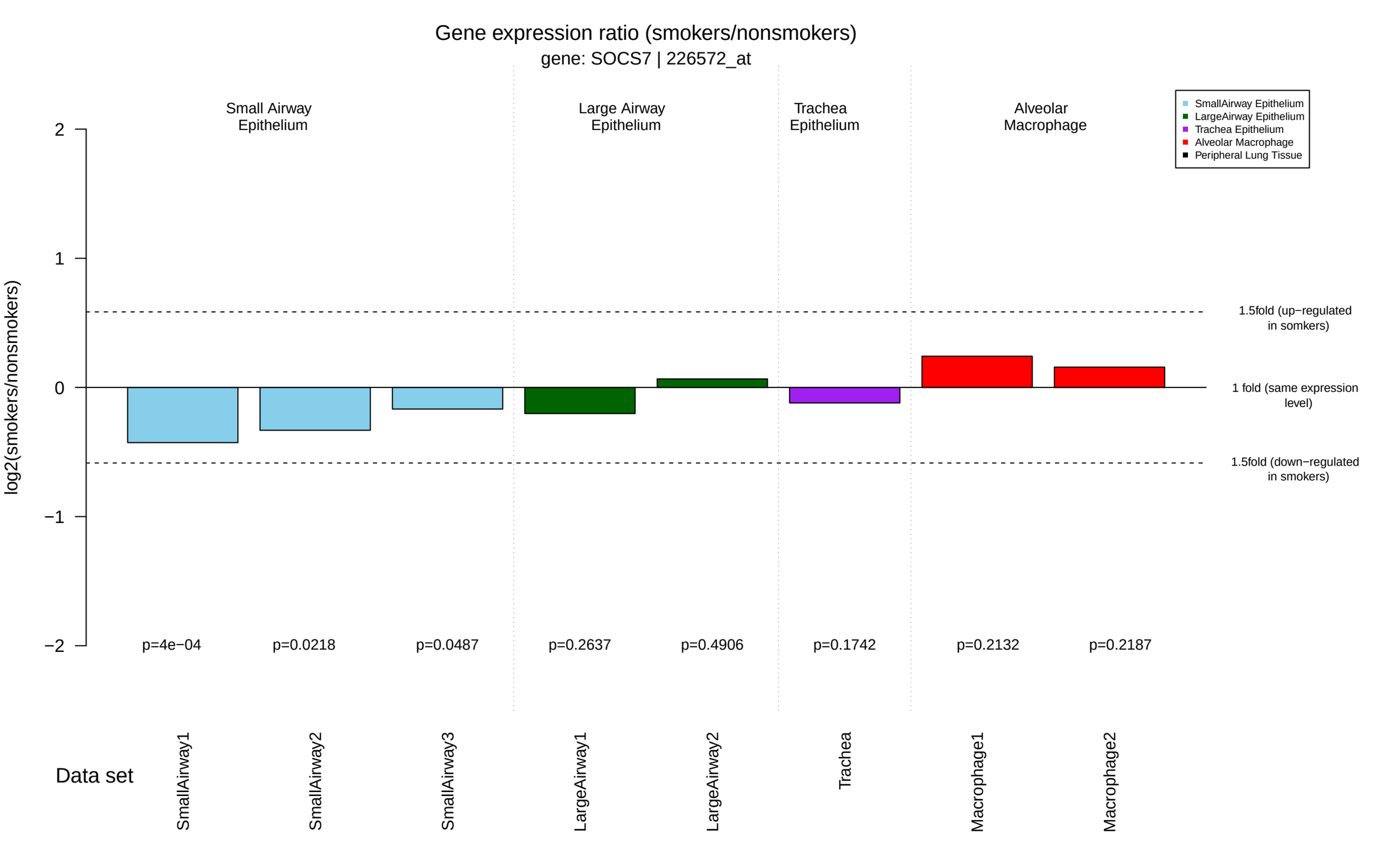
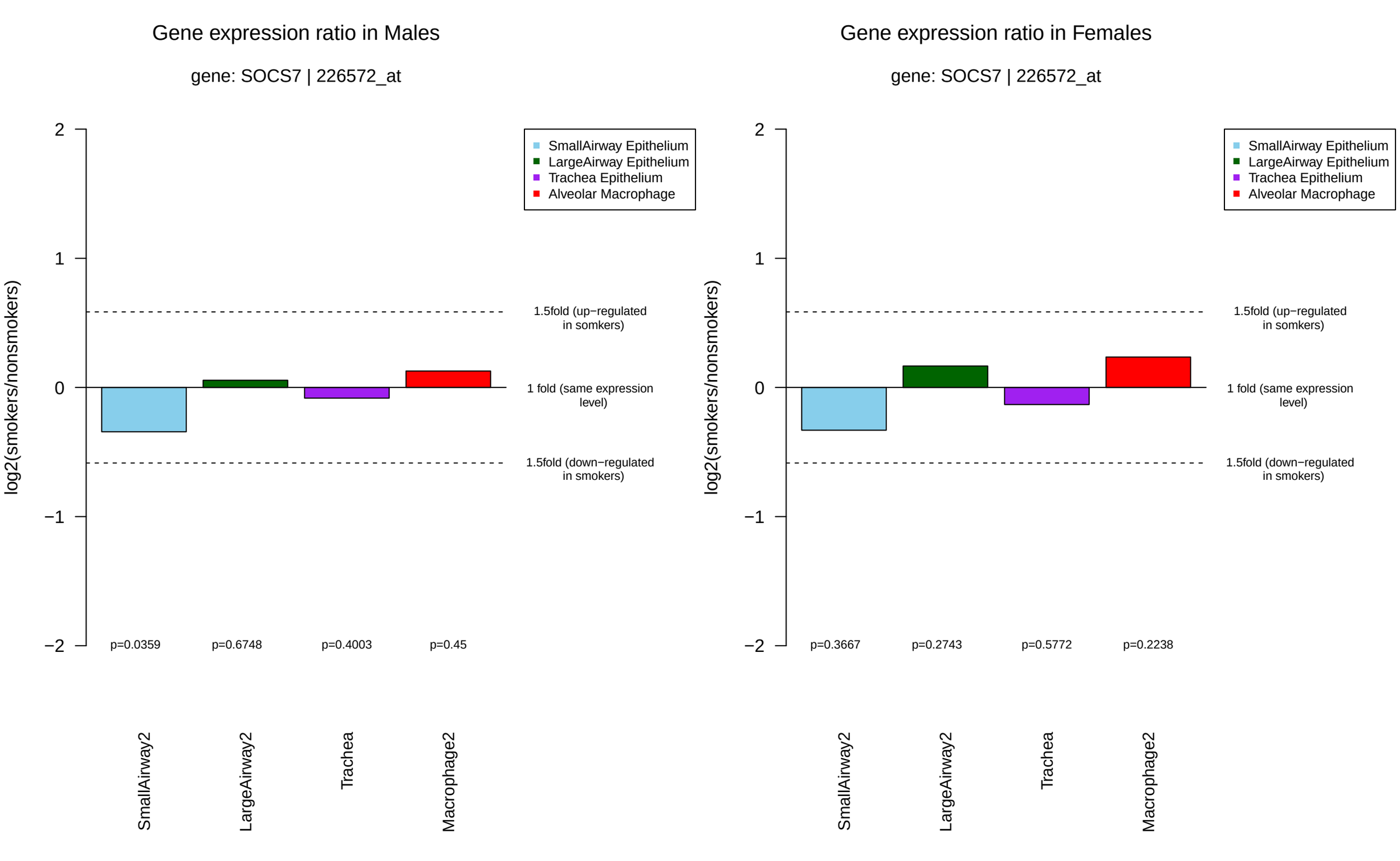
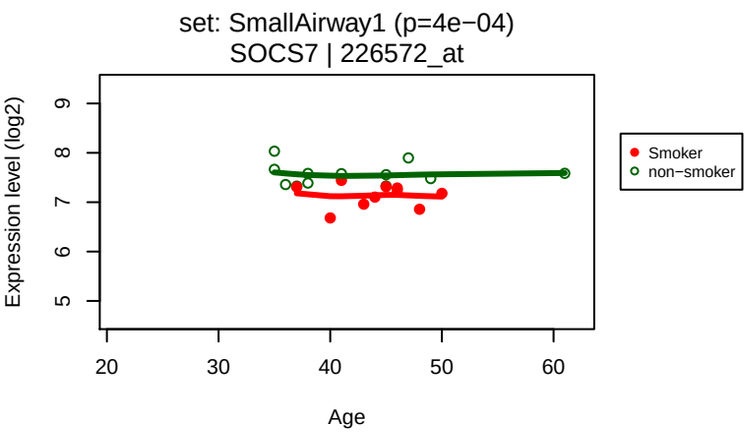
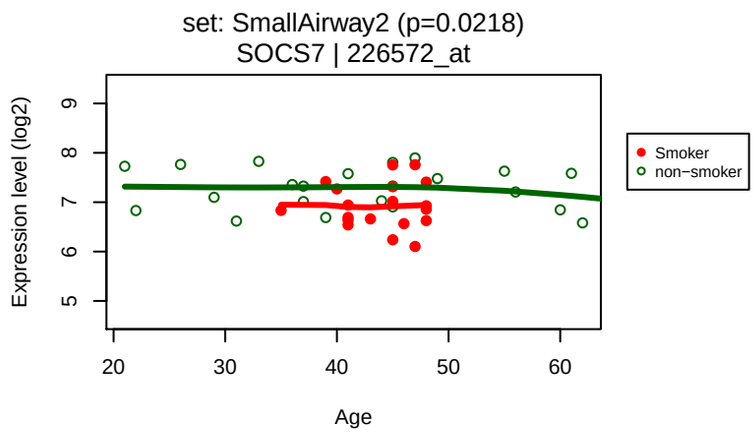
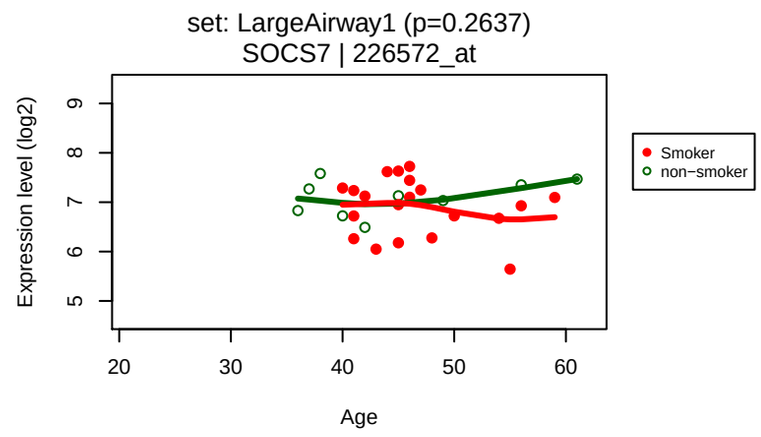

Gene expression of smokers and nonsmokers
gene: SOCS7 | 226572_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=4e−04
p=0.0218
p=0.0487
p=0.2637
p=0.4906
p=0.1742
p=0.2132
p=0.2187
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

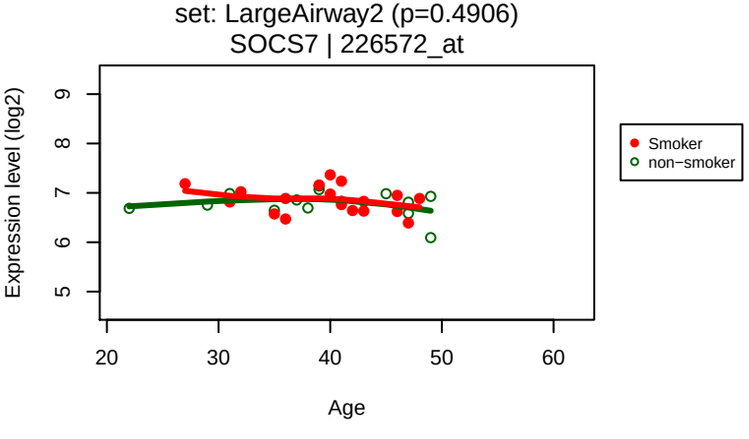

Gene expression ratio (smokers/nonsmokers)
gene: SOCS7 | 226572_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=4e−04
p=0.0218
p=0.0487
p=0.2637
p=0.4906
p=0.1742
p=0.2132
p=0.2187
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

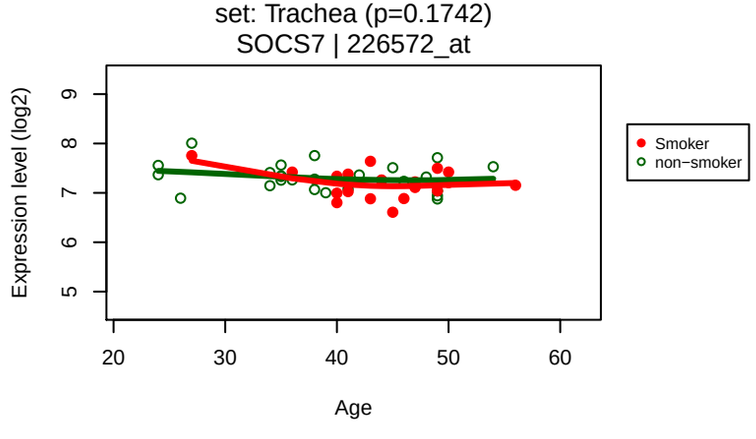

Gene expression ratio in Males
gene: SOCS7 | 226572_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0359
p=0.6748
p=0.4003
p=0.45
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SOCS7 | 226572_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3667
p=0.2743
p=0.5772
p=0.2238
SmallAirway2
LargeAirway2
Trachea
Macrophage2

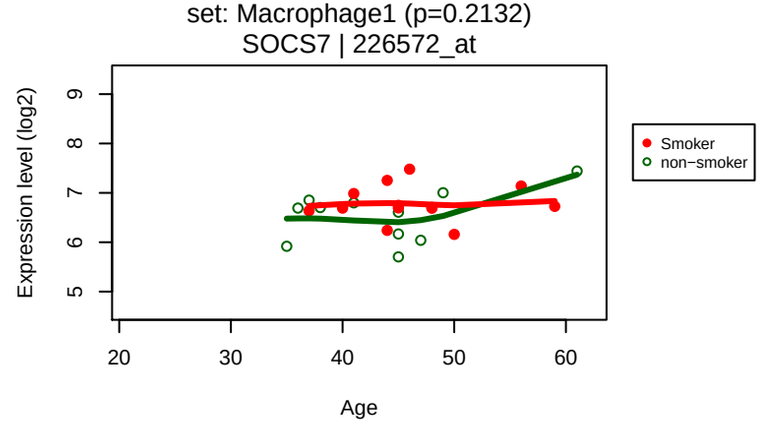

set: SmallAirway1 (p=4e−04)
SOCS7 | 226572_at
Expression level (log2)
Age
Smoker
non−smoker

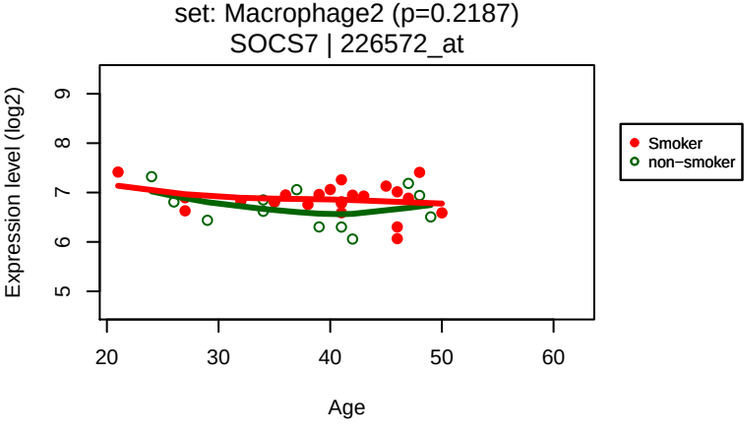

set: SmallAirway2 (p=0.0218)
SOCS7 | 226572_at
Expression level (log2)
Age
Smoker
non−smoker

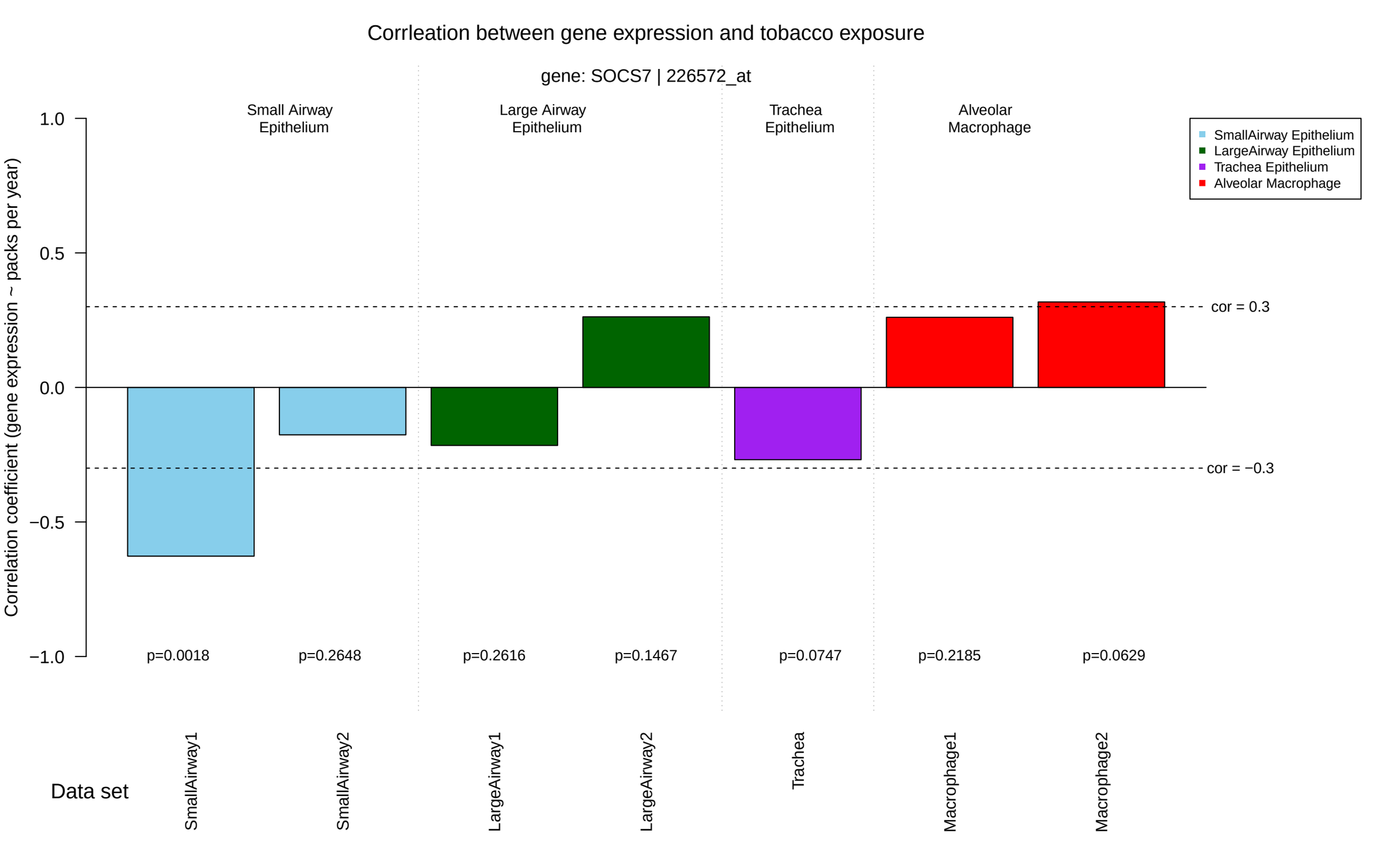

set: LargeAirway1 (p=0.2637)
SOCS7 | 226572_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4906)
SOCS7 | 226572_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1742)
SOCS7 | 226572_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2132)
SOCS7 | 226572_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2187)
SOCS7 | 226572_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SOCS7 | 226572_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0018
p=0.2648
p=0.2616
p=0.1467
p=0.0747
p=0.2185
p=0.0629
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2